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ULTRASONIC TREATMENT INSTRUMENT

Abstract

An ultrasonic treatment instrument includes: an ultrasonic blade configured to supply ultrasonic vibration and a high frequency current to a living tissue, respectively; a jaw configured to open and close with respect to the ultrasonic blade; a holder provided in the jaw to be swingable with respect to the jaw; and an abutment portion provided in the holder and made of a first resin material, the abutment portion being configured to abut against the ultrasonic blade when the jaw is closed with respect to the ultrasonic blade, at least a part of the holder being made of a second resin material, and the holder including an electrode configured to supply the high frequency current on at least a part of a contact surface that is included in the holder, the contact surface being configured to be in contact with the living tissue.

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Background/Summary

CROSS REFERENCES TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/JP2024/009171, filed on Mar. 8, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND
1. Technical Field
[0002] The present disclosure relates to an ultrasonic treatment instrument.
2. Related Art
[0003] In the related art, there has been known an ultrasonic treatment instrument that supplies ultrasonic vibration and a high frequency current as treatment energy to a site (hereinafter, referred to as a treatment target) to be treated in a living tissue to treat the treatment target (see, for example, US 2021/0196334 A).
[0004] The ultrasonic treatment instrument described in US 2021/0196334 A includes an ultrasonic blade, a jaw, and an abutment portion described below.
[0005] The ultrasonic blade supplies the ultrasonic vibration and the high frequency current to the treatment target, respectively.
[0006] The jaw opens and closes with respect to the ultrasonic blade. In addition, the jaw is provided with an electrode that supplies the high frequency current to the treatment target.
[0007] The abutment portion is made of a resin material and is provided on the jaw. The abutment portion abuts against the ultrasonic blade when the jaw is closed with respect to the ultrasonic blade.
SUMMARY
[0008] In some embodiments, an ultrasonic treatment instrument includes: an ultrasonic blade configured to supply ultrasonic vibration and a high frequency current to a living tissue, respectively; a jaw configured to open and close with respect to the ultrasonic blade; a holder provided in the jaw to be swingable with respect to the jaw; and an abutment portion provided in the holder and made of a first resin material, the abutment portion being configured to abut against the ultrasonic blade when the jaw is closed with respect to the ultrasonic blade, at least a part of the holder being made of a second resin material, and the holder including an electrode configured to supply the high frequency current on at least a part of a contact surface that is included in the holder, the contact surface being configured to be in contact with the living tissue.
[0009] The above and other features, advantages and technical and industrial significance of this invention will be better understood by reading the following detailed description of presently preferred embodiments of the disclosure, when considered in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0010] FIG. **1** is a diagram illustrating a treatment system according to an embodiment;
[0011] FIG. **2** is a diagram for describing a configuration of a distal end portion in an ultrasonic treatment instrument;
[0012] FIG. **3** is a diagram for describing a configuration of the distal end portion in the ultrasonic

treatment instrument;
[0013] FIG. **4** is a diagram for describing a configuration of a jaw;
[0014] FIG. **5** is a diagram for describing a configuration of a holder; and
[0015] FIG. **6** is a diagram for describing a modification of the embodiment.

DETAILED DESCRIPTION

[0016] Hereinafter, modes for carrying out the present invention (hereinafter, embodiments) will be described with reference to the drawings. The present invention is not limited by the embodiments described below. Furthermore, in the description of the drawings, the same portions are denoted by the same reference numerals.

Schematic Configuration of Treatment System

[0017] FIG. **1** is a diagram illustrating a treatment system **1** according to an embodiment.
[0018] The treatment system **1** applies treatment energy to a site to be treated (hereinafter, referred to as a treatment target) in a living tissue to treat the treatment target. The treatment energy in the present embodiment is ultrasonic energy and high frequency energy. In addition, the treatment that can be performed by the treatment system **1** according to the present embodiment is treatment such as coagulation (sealing) of the treatment target or incision of the treatment target. The coagulation and incision may be performed simultaneously. As illustrated in FIG. **1**, the treatment system **1** includes a treatment instrument **2** and a control device **3**.

Configuration of Treatment Instrument

[0019] Hereinafter, one side along a central axis Ax**1** (FIG. **1**) of an outer pipe **10** is referred to as a distal end side Ar**1**, and the other side is referred to as a proximal end side Ar**2**. In addition, a "width direction" described below is a direction orthogonal to the central axis Ax**1** and an opening/closing direction of a jaw **11** with respect to a treatment unit **131**, and means a direction orthogonal to a paper surface of FIG. **1**.
[0020] FIGS. **2** and **3** are diagrams for describing a configuration of a distal end portion in the treatment instrument **2**. Specifically, FIG. **2** is a perspective diagram illustrating the distal end portion of the treatment instrument **2**. FIG. **3** is a cross-sectional diagram of the distal end portion of the treatment instrument **2** taken along a plane orthogonal to the central axis Ax**1**.
[0021] The treatment instrument **2** is an ultrasonic treatment instrument. Then, the treatment instrument **2** treats a treatment target by applying ultrasonic energy and high frequency energy to the treatment target. As illustrated in FIG. **1**, the ultrasonic treatment instrument **2** includes a handpiece **4** and an ultrasonic transducer portion **5**.
[0022] As illustrated in FIGS. **1** to **3**, the handpiece **4** includes a fixed handle **6** (FIG. **1**), an operation handle **7** (FIG. **1**), a switch **8** (FIG. **1**), a rotary knob **9** (FIG. **1**), the outer pipe **10** (FIGS. **1** and **2**), the jaw **11**, a holder **14** (FIGS. **2** and **3**), an abutment portion **12** (FIG. **3**), and the ultrasonic blade **13**.
[0023] The fixed handle **6** is a portion that supports the entire treatment instrument **2** and is gripped by an operator (user) such as a surgeon.
[0024] The operation handle **7** is movably attached to the fixed handle **6** and receives an opening/closing operation by the operator such as a surgeon.
[0025] The switch **8** is provided in a state of being exposed to an outside of the fixed handle **6**, and receives a treatment operation by the operator such as a surgeon.
[0026] The rotary knob **9** has a substantially cylindrical shape coaxial with the central axis Ax**1**, and is provided on the distal end side Ar**1** of the fixed handle **6**. Then, the rotary knob **9** receives a rotation operation by the operator such as a surgeon. By the rotation operation, the rotary knob **9** rotates about the central axis Ax**1** with respect to the fixed handle **6**. The outer pipe **10**, the jaw **11**, the holder **14**, the abutment portion **12**, and the ultrasonic blade **13** rotate about the central axis Ax**1** by the rotation of the rotary knob **9**.
[0027] The outer pipe **10** has a tubular shape and corresponds to a pipe. In the present embodiment, the outer pipe **10** is a cylindrical pipe made of an electrically conductive material such as metal.

[0028] In the outer pipe **10**, a first pin Pi**1** (FIGS. **1** and **2**) having a cylindrical shape extending in a direction orthogonal to the paper surface of FIG. **1** and engaging with the jaw **11** and pivotally supporting the jaw **11** in a rotatable manner is fixed to an end portion on the distal end side Ar**1**. In the present embodiment, the first pin Pi**1** is made of an electrically conductive material such as metal.

[0029] An outer peripheral surface of the outer pipe **10** is covered with an electrically insulating outer tube (not illustrated). Further, a tubular inner pipe PI (FIG. **2**) that moves forward and backward along a longitudinal direction of the outer pipe **10** according to the opening/closing operation to the operation handle **7** by the operator such as a surgeon is inserted into the outer pipe **10**. A second pin Pi**2** (FIG. **2**) having a cylindrical shape extending in a direction orthogonal to the paper surface of FIG. **1** and engaging with the jaw **11** is fixed to an end portion of the inner pipe PI on the distal end side Ar**1**. In the present embodiment, the second pin Pi**2** is disposed on an upper side (a side on which a jaw main body **111** is disposed with respect to the treatment unit **131**) in FIG. **2** with respect to the first pin Pi**1**.

[0030] The jaw **11** is coupled to the outer pipe **10** by the first pin Pi**1**. The jaw **11** is coupled to the inner pipe PI by the second pin Pi**2**. Then, the jaw **11** rotates about the first pin Pi**1** with respect to the outer pipe **10** in conjunction with the forward and backward movement of the inner pipe PI according to the opening/closing operation to the operation handle **7** by the operator such as a surgeon. As a result, the jaw **11** is opened and closed with respect to the treatment unit **131** which is the end portion of the ultrasonic blade **13** on the distal end side, and can grip a treatment target with the treatment unit **131** therebetween.

[0031] Note that the treatment instrument **2** may be a push-close type or a pull-close type.

[0032] The push-close type has the following configuration.

[0033] The jaw **11** rotates about the first pin Pi**1** in a direction approaching the treatment unit **131** in conjunction with the movement of the inner pipe PI toward the distal end side Ar**1**. That is, the jaw **11** is closed with respect to the treatment unit **131**. The jaw **11** rotates about the first pin Pi**1** in a direction away from the treatment unit **131** in conjunction with the movement of the inner pipe PI toward the proximal end side Ar**2**. That is, the jaw **11** opens with respect to the treatment unit **131**.

[0034] The pull-close type has the following configuration.

[0035] The jaw **11** rotates about the first pin Pi**1** in the direction approaching the treatment unit **131** in conjunction with the movement of the inner pipe PI toward the proximal end side Ar**2**. That is, the jaw **11** is closed with respect to the treatment unit **131**. The jaw **11** rotates about the first pin Pi**1** in a direction away from the treatment unit **131** in conjunction with the movement of the inner pipe PI toward the distal end side Ar**1**. That is, the jaw **11** opens with respect to the treatment unit **131**.

[0036] A detailed configuration of the jaw **11** will be described in “Configuration of Jaw” described later.

[0037] The holder **14** has a cylindrical shape extending in a direction orthogonal to the paper surface of FIG. **1**, and is pivotally supported with respect to the jaw **11** so as to be swingable about the central axis of a third pin Pi**3** by the third pin Pi**3** (FIG. **2**) fixed to the jaw **11**. That is, by making the holder **14** swingable about the central axis of the third pin Pi**3**, a position where a strongest force is applied to a treatment target when the treatment target is gripped between the jaw **11** and the treatment unit **131** is positioned at substantially the center of the jaw **11** in the longitudinal direction instead of the proximal end side Ar**2** of the jaw **11**. That is, a force is substantially uniformly applied to the treatment target gripped between the jaw **11** and the treatment unit **131**. In the present embodiment, the third pin Pi**3** is made of an electrically conductive material such as metal.

[0038] The holder **14** is provided with an electrode EP and first and second electrically conductive surfaces CS**1** and CS**2** (see FIG. **5**).

[0039] A detailed configuration of the holder **14** will be described in “Configuration of Holder” described later. The configurations of the electrode EP and the first and second electrically

conductive surfaces CS**1** and CS**2** will be described in "Configurations of Electrode and First and Second Electrically conductive Surfaces" described later.
[0040] The abutment portion **12** is made of a first resin material having electrical insulation properties and biocompatibility, for example, polytetrafluoroethylene (PTFE), and has a substantially rectangular parallelepiped shape extending along the longitudinal direction of the jaw **11** and the holder **14**. Then, as illustrated in FIGS. **2** and **3**, the abutment portion **12** is fixed to a surface of a holder main body **141** on a side of the treatment unit **131**, and abuts against the treatment unit **131** when the jaw **11** is closed with respect to the treatment unit **131**. The abutment portion **12** has a function of preventing the treatment unit **131** which is ultrasonically vibrated from being damaged by colliding with the jaw **11** when the incision of the treatment target by ultrasonic vibration is completed.
[0041] The ultrasonic blade **13** is made of an electrically conductive material and has an elongated shape extending along the central axis Ax**1**. In addition, the ultrasonic blade **13** is inserted into the inner pipe PI in a state where the treatment unit **131** protrudes to the outside. At this time, an end portion of the ultrasonic blade **13** on the proximal end side Ar**2** is mechanically coupled to an ultrasonic transducer **52** configuring the ultrasonic transducer portion **5** as illustrated in FIG. **1**. Then, the ultrasonic blade **13** transmits ultrasonic vibration generated by the ultrasonic transducer portion **5** from the end portion on the proximal end side Ar**2** to the treatment unit **131**. The ultrasonic vibration is longitudinal vibration that vibrates in a direction along the central axis Ax**1**. In the ultrasonic blade **13**, an outer peripheral surface excluding the treatment unit **131** is covered with an electrically insulating inner tube.
[0042] As illustrated in FIG. **1**, the ultrasonic transducer portion **5** includes a transducer (TD) case **51** and the ultrasonic transducer **52**.
[0043] The TD case **51** supports the ultrasonic transducer **52** and is detachably coupled to the fixed handle **6**.
[0044] The ultrasonic transducer **52** generates ultrasonic vibration under the control of the control device **3**. In the present embodiment, the ultrasonic transducer **52** is configured by a bolted Langevin transducer (BLT).
Configuration of Control Device
[0045] The control device **3** integrally controls an operation of the treatment instrument **2** via an electric cable C (FIG. **1**).
[0046] Specifically, the control device **3** detects a treatment operation on the switch **8** by the operator such as a surgeon via the electric cable C. Then, when detecting the treatment operation, the control device **3** applies treatment energy to the treatment target gripped between the jaw **11** and the treatment unit **131** via the electric cable C. That is, the control device **3** treats the treatment target.
[0047] For example, when applying ultrasonic energy to the treatment target, the control device **3** supplies drive power to the ultrasonic transducer **52** via the electric cable C. As a result, the ultrasonic transducer **52** generates longitudinal vibration (ultrasonic vibration) that vibrates in a direction along the central axis Ax**1**. In addition, the treatment unit **131** vibrates with a desired amplitude by the longitudinal vibration. Then, the ultrasonic vibration is supplied from the treatment unit **131** to the treatment target gripped between the jaw **11** and the treatment unit **131**. The ultrasonic energy is applied from the treatment unit **131** to the treatment target.
[0048] Furthermore, for example, when applying high frequency energy to the treatment target, the control device **3** supplies high frequency power between the electrode EP provided on the holder **14** and the ultrasonic blade **13** via the electric cable C or the like. When the high frequency power is supplied between the electrode EP and the ultrasonic blade **13**, a high frequency current is supplied to the treatment target gripped between the jaw **11** and the treatment unit **131**. In other words, the high frequency energy is applied to the treatment target.
Configuration of Jaw

[0049] FIG. **4** is a diagram for describing a configuration of the jaw **11**. Specifically, FIG. **4** is a perspective diagram of the jaw **11** as viewed from the side of the treatment unit **131**. In FIG. **4**, illustration of the distal end portion is omitted for convenience of description.
[0050] The jaw **11** is made of an electrically conductive material. As illustrated in FIG. **4**, the jaw **11** is a member in which a jaw main body **111** and a pair of bearing portions **112** are integrally formed.
[0051] The jaw main body **111** is formed of a substantially elongated plate body.
[0052] In the jaw main body **111**, as illustrated in FIG. **4**, a recess **1111** extending from the proximal end toward the distal end side Ar**1** along the longitudinal direction of the jaw main body **111** is provided on a surface on the side of the treatment unit **131**.
[0053] In side wall portions on both sides in the width direction of the jaw main body **111** configuring the recess **1111**, as illustrated in FIG. **4**, through-holes **1113** penetrating the front and back of the side wall portions are provided at positions substantially at the center of the jaw **11** in the longitudinal direction. The third pin Pi**3** inserted into the through-holes **1113** is fixed by welding.
[0054] In addition, a cover RC (FIGS. **2** to **4**) made of an electrically insulating resin is integrally formed on a back side surface of the jaw main body **111** separated from the treatment unit **131** in a state of covering the back side surface. In the present embodiment, the cover RC is insert-molded with respect to the jaw main body **111**, but the disclosure is not limited thereto. For example, a configuration in which the cover RC is fixed to the jaw main body **111** by snap-fitting or a metal pin may be adopted.
[0055] The pair of bearing portions **112** are provided at end portions of the jaw main body **111** on the proximal end side Ar**2**, and are configured by plate bodies facing each other in the width direction of the jaw main body **111**. The pair of bearing portions **112** have the same configuration. Therefore, only the configuration of one bearing portion **112** will be described below.
[0056] The bearing portion **112** is provided with first and second insertion holes **1121** and **1122** penetrating the front and back, respectively. The bearing portion **112** is coupled to the outer pipe **10** by inserting the first pin Pi**1** into the first insertion hole **1121**. The bearing portion **112** is coupled to the inner pipe PI by inserting the second pin Pi**2** into the second insertion hole **1122**.
Configuration of Holder **14**
[0057] FIG. **5** is a diagram for describing a configuration of the holder **14**. Specifically, FIG. **5** is a perspective diagram of the holder **14** as viewed from the side of the treatment unit **131**. In FIG. **5**, dots are added to the electrode EP and the first and second electrically conductive surfaces CS**1** and CS**2** for convenience of description.
[0058] The holder **14** is made of a second resin material having electrical insulation properties and biocompatibility, for example, polyether ether ketone (PEEK) or polyphenyl sulfone (PPSU). As illustrated in FIG. **5**, the holder **14** is a member in which a holder main body **141**, a plurality of first tooth portions **142**, and a plurality of second tooth portions **143** are integrally formed.
[0059] The holder main body **141** is formed of an elongated plate body. In addition, an outer shape of the holder main body **141** is set to be substantially the same as an inner surface shape of the recess **1111**.
[0060] As illustrated in FIG. **5**, the plurality of first tooth portions **142** respectively protrude from one side in the width direction of the surface of the holder main body **141** on the side of the treatment unit **131** toward the side of the treatment unit **131**, and are arranged side by side along the longitudinal direction of the holder main body **141**.
[0061] As illustrated in FIG. **5**, the plurality of second tooth portions **143** respectively protrude from the other side in the width direction of the surface of the holder main body **141** on the side of the treatment unit **131** toward the side of the treatment unit **131**, and are arranged side by side along the longitudinal direction of the holder main body **141**.
[0062] Here, on the surface of the holder main body **141** on the side of the treatment unit **131**, at

the center portion in the width direction located between the plurality of first tooth portions **142** and the plurality of second tooth portions **143**, as illustrated in FIG. **5**, a recess **144** that is recessed on a side separated from the treatment unit **131** and extending along the longitudinal direction of the holder main body **141** is provided. Further, claw portions **145** and **146** (FIGS. **3** and **5**) protruding toward the center in the width direction and extending along the longitudinal direction of the holder main body **141** are respectively provided in side wall portions on both sides in the width direction of the holder main body **141** configuring the recess **144**. Then, the abutment portion **12** is mechanically fixed to the holder **14** by being locked to the claw portions **145** and **146**. That is, the abutment portion **12** is provided at the center portion of the holder **14** in the width direction. Further, in the side wall portions on both sides in the width direction of the holder main body **141** configuring the recess **144**, insertion holes **147** which respectively penetrate through the front and back of the side wall portions and through which the third pin Pi**3** is inserted are provided.

Configurations of Electrode and First and Second Electrically Conductive Surfaces

[0063] As illustrated in FIG. **5**, the electrode EP is provided on each of inner surfaces of the plurality of first tooth portions **142** in the width direction (on the side of the second tooth portion **143**), distal end surfaces of the first tooth portions **142**, inner surfaces of the plurality of second tooth portions **143** in the width direction (the side of the first tooth portion **142**), and distal end surfaces of the second tooth portions **143**, respectively. In addition, a coating material having non-adhesiveness to the treatment target is applied onto the electrode EP. The coating material is an extremely thin coating material including fluorine or silicon and having a thickness of about several hundred nm to several μm.

[0064] As illustrated in FIG. **5**, the first electrically conductive surface CS**1** is provided on an inner surface of the insertion hole **147**.

[0065] As illustrated in FIG. **5**, the second electrically conductive surface CS**2** is provided on the side wall portion of the recess **144**, and electrically couples the electrode EP and the first electrically conductive surface CS**1**.

[0066] The electrode EP and the first and second electrically conductive surfaces CS**1** and CS**2** described above are formed by the following three-dimensional plating processing.

[0067] Specifically, the holder **14** made of the second resin material is irradiated with a laser at a predetermined position. Thereafter, the electrode EP and the first and second electrically conductive surfaces CS**1** and CS**2** are formed at the laser irradiation position by electroless plating. A thickness of the electrode EP and the first and second electrically conductive surfaces CS**1** and CS**2** is, for example, about several μm. In addition, a formed product produced by such three-dimensional plating processing is called a molded interconnect device (MID) as a product in which an electrode circuit is formed on the surface of a three-dimensional resin molded article. The electrode EP and the first and second electrically conductive surfaces CS**1** and CS**2** are not limited to those formed by the above-described three-dimensional plating processing, and those formed by other methods may be adopted.

[0068] Then, when high frequency power is supplied to the electrode EP in order to apply high frequency energy to the treatment target, the high frequency power is supplied following an electric path of the electric cable C to the outer pipe **10** to the first pin Pi**1** to the jaw **11** to the third pin Pi**3** to the first electrically conductive surface CS**1** to the second electrically conductive surface CS**2** and to the electrode EP.

[0069] According to the present embodiment described above, the following effects are obtained.

[0070] In the treatment instrument **2** according to the present embodiment, at least a part of the holder **14** is made of the second resin material. The holder **14** is provided with the electrode EP described above. Therefore, rigidity of the holder **14** can be made smaller than that in a case where it is made of a metal material. That is, even when the abutment portion **12** is worn by repeatedly outputting the ultrasonic vibration, a distance between the ultrasonic blade **13** and the holder **14** decreases, and the ultrasonic blade **13** and the holder **14** come into contact with each other, an

influence on the ultrasonic blade **13** can be reduced.
[0071] Therefore, the treatment instrument **2** according to the present embodiment can reduce an influence on a life of the ultrasonic blade **13**.
[0072] In particular, the electrode EP is formed by three-dimensional plating processing. Therefore, even when the holder **14** is made of the second resin material, the electrode EP can be easily formed at a specific position in the holder **14**.
[0073] In addition, a coating material having non-adhesiveness to the treatment target is provided on the electrode EP. Therefore, it is possible to prevent the treatment target from adhering to the electrode EP and to favorably treat the treatment target.
Other Embodiments
[0074] Although the embodiment for carrying out the present invention have been described so far, the present invention should not be limited only by the above-described embodiment.
[0075] In the above-described embodiment, the outer pipe **10** is adopted as the electric path from the electric cable C to the electrode EP, but the present invention is not limited thereto, and the inner pipe PI may be adopted instead of the outer pipe **10**.
[0076] FIG. **6** is a diagram for describing a modification of the embodiment. Specifically, FIG. **6** is a diagram corresponding to FIG. **4**. In FIG. **6**, dots are added to a third electrically conductive surface CS**3** for convenience of description.
[0077] In the above-described embodiment, the jaw **11** is made of an electrically conductive material, but the present invention is not limited thereto, and may be made of a third resin material. As the third resin material, a resin material similar to the second resin material described in the above-described embodiment can be adopted. In this case, as illustrated in FIG. **6**, the jaw **11** is provided with the third electrically conductive surface CS**3** in order to secure the electric path from the first pin Pi**1** to the third pin Pi**3**.
[0078] The third electrically conductive surface CS**3** is provided on the inner surface of the first insertion hole **1121**, the inner surface of the through-hole **1113**, the surface of the bearing portion **112** facing the other bearing portion **112**, and the side wall portion of the recess **1111**, and secures an electric path from the first pin Pi**1** inserted into the first insertion hole **1121** to the third pin Pi**3** inserted into the through-hole **1113**.
[0079] The third electrically conductive surface CS**3** described above is formed by three-dimensional plating processing similarly to the electrode EP and the first and second electrically conductive surfaces CS**1** and CS**2** described in the above-described embodiment. The third electrically conductive surface CS**3** is not limited to one formed by the above-described three-dimensional plating processing, and one formed by another method may be adopted.
[0080] Even in a case where the configuration of the present modification illustrated in FIG. **6** is adopted, the same effects as those of the above-described embodiment are obtained.
[0081] According to an ultrasonic treatment instrument of the disclosure, an influence on a life of an ultrasonic blade can be reduced.
[0082] Additional advantages and modifications will readily occur to those skilled in the art. Therefore, the disclosure in its broader aspects is not limited to the specific details and representative embodiments shown and described herein. Accordingly, various modifications may be made without departing from the spirit or scope of the general inventive concept as defined by the appended claims and their equivalents.

Claims

1. An ultrasonic treatment instrument comprising: an ultrasonic blade configured to supply ultrasonic vibration and a high frequency current to a living tissue, respectively; a jaw configured to open and close with respect to the ultrasonic blade; a holder provided in the jaw to be swingable with respect to the jaw; and an abutment portion provided in the holder and made of a first resin material, the abutment portion being configured to abut against the ultrasonic blade when the jaw is closed with respect to the ultrasonic blade, at least a part of the holder being made of a second resin material, and the holder including an electrode configured to supply the high frequency current on at least a part of a contact surface that is included in the holder, the contact surface being configured to be in contact with the living tissue.

2. The ultrasonic treatment instrument according to claim 1, wherein the abutment portion is provided at a center portion of the holder in a width direction, and the electrode is provided on at least a part of each of contact surfaces located on both sides of the abutment portion in the width direction.

3. The ultrasonic treatment instrument according to claim 1, wherein the second resin material is polyether ether ketone or polyphenyl sulfone.

4. The ultrasonic treatment instrument according to claim 1, wherein the electrode is formed by three-dimensional plating processing.

5. The ultrasonic treatment instrument according to claim 1, further comprising: a cylindrical pin attached to the jaw and swingably supporting the holder, wherein the electrode is electrically coupled to the jaw via the pin.

6. The ultrasonic treatment instrument according to claim 5, wherein the holder is provided with an insertion hole through which the pin is inserted, and an inner surface of the insertion hole is provided with a first electrically conductive surface configured to electrically couple the pin and the electrode.

7. The ultrasonic treatment instrument according to claim 6, wherein the holder is provided with a second electrically conductive surface configured to electrically couple the first electrically conductive surface and the electrode.

8. The ultrasonic treatment instrument according to claim 1, wherein a coating material having non-adhesiveness to the living tissue is provided on the electrode.

9. The ultrasonic treatment instrument according to claim 6, wherein the jaw is made of a third resin material and includes a third electrically conductive surface configured to be electrically coupled to the pin.

10. The ultrasonic treatment instrument according to claim 6, wherein the first resin material is polytetrafluoroethylene (PTFE), the second resin material is polyether ether ketone (PEEK) or polyphenyl sulfone (PPSU), and the third resin material is polyether ether ketone (PEEK) or polyphenyl sulfone (PPSU).
